

Chapter 24: Proximity Operators

Convex Analysis and Monotone Operator Theory

Based on Bauschke & Combettes (2017)

Overview I

- 1 Introduction to Proximity Operators
- 2 Basic Properties of Proximity Operators
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What is a Proximity Operator?

Definition

For a function $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \mathbb{R}_{++}$, the **proximity operator** is:

$$\text{Prox}_{\gamma f}(x) := \underset{p \in \mathcal{H}}{\text{argmin}} \left(f(p) + \frac{1}{2\gamma} \|p - x\|^2 \right)$$

- f : proper, convex, lower-semicontinuous function
- $\gamma > 0$: step size or regularization parameter
- The argmin is **unique** (strong convexity)

Interpretation

Balances two goals:

- 1 Minimize $f(p)$
- 2 Stay close to the original point x

Basic Characterization

Theorem (Characterization via Subdifferential)

For $f \in \Gamma_0(\mathcal{H})$, $x \in \mathcal{H}$, and $\gamma \in \mathbb{R}_{++}$:

$$p = \text{Prox}_{\gamma f}(x) \iff 0 \in \partial f(p) + \frac{1}{\gamma}(p - x)$$

Corollary

Equivalently:

$$p = \text{Prox}_{\gamma f}(x) \iff x - p \in \gamma \partial f(p)$$

Example

If f is differentiable at p :

$$x - p = \gamma \nabla f(p) \implies p = x - \gamma \nabla f(p)$$

This is a **gradient descent step**!

Proximity Operators as Resolvents

Theorem (Connection to Monotone Operators)

For $A = \partial f$ (maximally monotone by Theorem 20.25):

$$\text{Prox}_{\gamma f} = (\text{Id} + \gamma A)^{-1}$$

- $\text{Prox}_{\gamma f}$ is the **resolvent** of the monotone operator ∂f
- Key property: For all $f \in \Gamma_0(\mathcal{H})$:

$$J_{\gamma \partial f} = \text{Prox}_{\gamma f}$$

- Single-valued and nonexpansive operator

Corollary (Relationship to Zeros)

Finding p such that $0 \in \partial f(p)$ is equivalent to finding fixed points of $\text{Prox}_{\gamma f}$.

Proximity Operators and Gradients

Theorem (Characterization via Gradients)

Let $T : \mathcal{H} \rightarrow \mathcal{H}$, $f \in \Gamma_0(\mathcal{H})$, and $q = \frac{1}{2} \|\cdot\|^2$. Then:

- ① $T = \text{Prox}_f$
- ② $T = \nabla(f + q)^*$
- ③ $T = \nabla(f^* \square q) = \nabla(! (f^*))$

are equivalent.

Corollary

For $f, g \in \Gamma_0(\mathcal{H})$ with $f = g - \frac{1}{2} \|\cdot\|^2$:

$$\text{Prox}_f = \nabla g^*$$

Key Properties I

[Composition Rules] Let $f, g \in \Gamma_0(\mathcal{H})$. We have:

- ① If $g = f + \frac{\alpha}{2} \|\cdot - z\|^2 + \langle u, \cdot \rangle + \beta$:

$$\text{Prox}_g x = \text{Prox}_{(1+\alpha)^{-1}f} \left((1+\alpha)^{-1}(x + \alpha z - u) \right)$$

- ② If $g = z + \text{Prox}_{Lf}(x - z)$ where L is invertible:

$$\text{Prox}_g x = L^*(\text{Prox}_{f \circ L}(Lx))$$

- ③ If $\text{Prox}_f = \text{Prox}_g$: There exists $\gamma \in \mathbb{R}$ such that $f = g + \gamma$.

Key Properties II

[Scaling and Operations]

- ① Scaling: $\text{Prox}_{\mu f}x = x + \frac{\gamma}{\mu}(\text{Prox}_{\gamma\mu f}x - x)$
- ② Reflection: If f is even, then Prox_f is odd and $\text{Prox}_f 0 = 0$
- ③ Conjugate: For f^* , $\text{Prox}_{f^*}x = x - \gamma\text{Prox}_{\gamma^{-1}f}(\gamma^{-1}x)$

Weak Sequential Continuity

Theorem (Weak Sequential Continuity)

For $f \in \Gamma_0(\mathcal{H})$:

- 1 Prox_f is weakly sequentially continuous
- 2 If $(x_n)_{n \in \mathbb{N}}$ is a sequence with $x_n \rightharpoonup x$, then:

$$\|\text{Prox}_f x_n\| \leq \|x_n\|$$

- 3 Therefore $(\text{Prox}_f x_n)_{n \in \mathbb{N}}$ is bounded, and $\text{Prox}_f x_n \rightharpoonup \text{Prox}_f x$

Corollary

Proximity operators are **continuous** (strong topology) and have strong convergence properties.

Composition of Proximity Operators

Theorem (Composition Rule)

Let $f, g \in \Gamma_0(\mathcal{H})$. If one of the following holds:

- ① $(\forall y \in \text{dom} \partial g) \partial g(y) \subseteq \partial(\text{Prox}_f y)$
- ② $(\forall (x, u) \in \text{gra} \partial f) \partial g(x + u) \subseteq \partial(x)$
- ③ $\text{dom} g$ is open, g is Gâteaux differentiable on $\text{dom} g$, $\text{dom} \partial f \subset \text{dom} g$, and $(\forall y \in \text{dom} g) \nabla g(y) = \nabla g(\text{Prox}_f y)$
- ④ $0 \in \text{sri}(\text{dom} f - \text{dom} g)$ and $(\forall (x, u) \in \text{gra} \partial f) \partial g(x) \subseteq \partial(x + u)$

Then:

$$\text{Prox}_{f+g} = \text{Prox}_f \circ \text{Prox}_g$$

Example 1: Quadratic Functions

Example (Self-Adjoint and Monotone Operator)

Let $L \in \mathcal{B}(\mathcal{H})$ be self-adjoint and monotone, $u \in \mathcal{H}$, $\alpha \in \mathbb{R}$. Set:

$$f : \mathcal{H} \rightarrow \mathbb{R}, \quad x \mapsto \frac{1}{2} \langle Lx | x \rangle + \langle u | x \rangle + \alpha$$

Then $f \in \Gamma_0(\mathcal{H})$ and:

$$(\forall x \in \mathcal{H}) \operatorname{Prox}_f x = (\operatorname{Id} + L)^{-1} (x - u)$$

Example (Weighted Norm Squared)

Let I be a finite index set, $(\mathcal{K}_i)_i$, $(r_i)_i$, $(\lambda_i)_i$ given. Set:

$$f(x) = \frac{1}{2} \sum_{i \in I} \lambda_i \|L_i^* x - r_i\|^2$$

Then:

$$(\forall x \in \mathcal{H}) \operatorname{Prox}_f x = \left(\operatorname{Id} + \sum_{i \in I} \lambda_i L_i^* L_i \right)^{-1} \left(\sum_{i \in I} \lambda_i L_i^* r_i \right)$$

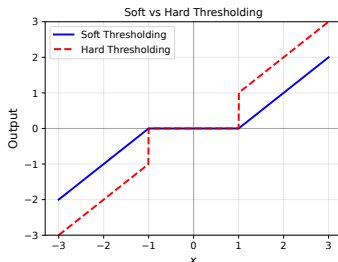
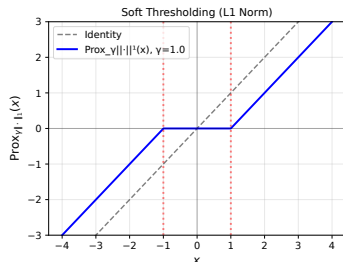
Example 2: Soft Thresholding

Example (Soft Thresholding for ℓ_1 Norm)

For $\gamma \in \mathbb{R}_{++}$ and $x \in \mathcal{H}$:

$$\text{Prox}_{\gamma\|\cdot\|_1} x = \begin{cases} \left(1 - \frac{\gamma}{\|x\|}\right) x & \text{if } \|x\| > \gamma \\ 0 & \text{if } \|x\| \leq \gamma \end{cases}$$

This is also called **soft thresholding** or **shrinkage** operator.



Example 3: Huber Function

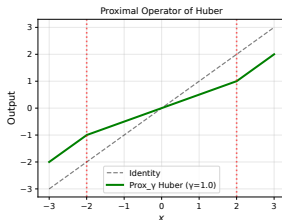
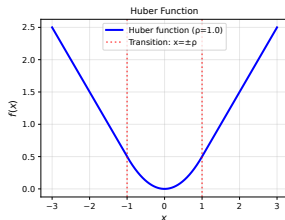
Example (Generalized Huber Function)

The function:

$$f : \mathcal{H} \rightarrow \mathbb{R}, \quad x \mapsto \begin{cases} \rho \|x\| - \frac{\rho^2}{2} & \text{if } \|x\| > \rho \\ \frac{\|x\|^2}{2} & \text{if } \|x\| \leq \rho \end{cases}$$

with $\gamma \in \mathbb{R}_{++}$ gives:

$$(\forall x \in \mathcal{H}) \operatorname{Prox}_{\gamma f} x = \begin{cases} \left(1 - \frac{\gamma \rho}{\|x\|}\right) x & \text{if } \|x\| > (\gamma + 1)\rho \\ \frac{1}{\gamma + 1} x & \text{if } \|x\| \leq (\gamma + 1)\rho \end{cases}$$



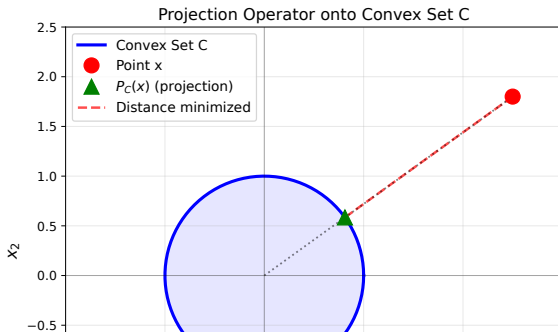
Example 4: Indicator Functions

Example (Projection onto Convex Set)

Let C be a nonempty closed convex subset of $\mathcal{H}$. The indicator function ι_C satisfies:

$$\text{Prox}_{\iota_C} = P_C$$

i.e., the proximity operator is the **projection** onto C .



Distance Functions in Optimization

Definition (Distance Function)

For a nonempty closed convex set $C \subseteq \mathcal{H}$:

$$d_C(x) := \inf_{p \in C} \|x - p\|$$

Theorem (Subdifferential of Distance Function)

For $x \notin C$:

$$\partial d_C(x) = \begin{cases} \{(P_C x - x) / \|x - P_C x\|\} & \text{if } d_C(x) > \max \text{Argmin} \phi; \\ P_C x & \text{if } d_C(x) \leq \max \text{Argmin} \phi \end{cases}$$

Corollary

The proximity operator of $f = \phi \circ d_C$ depends on whether the distance to C exceeds a threshold.

Proximal Operators of Distance Compositions

[Distance-Based Composition] Let C be a nonempty closed convex set, $\phi : \mathbb{R} \rightarrow \mathbb{R}$ even and differentiable on $\mathbb{R} \setminus \{0\}$, and $f = \phi \circ d_C$. Then:

$$\text{Prox}_f x = \begin{cases} x + \frac{\text{Prox}_\phi d_C(x)}{d_C(x)} (P_C x - x) & \text{if } d_C(x) > \max \partial \phi(0) \\ P_C x & \text{if } d_C(x) \leq \max \partial \phi(0) \end{cases}$$

Example (Weighted Distance to Convex Set)

Let C be nonempty closed convex, $\gamma \in \mathbb{R}_{++}$, $x \in \mathcal{H}$:

$$\text{Prox}_{\gamma d_C} x = \begin{cases} x + \frac{\gamma}{d_C(x)} (P_C x - x) & \text{if } d_C(x) > \gamma \\ P_C x & \text{if } d_C(x) \leq \gamma \end{cases}$$

Proximity Operators on $\mathbb{R}$

Theorem (Increasing Property)

Let $f \in \Gamma_0(\mathbb{R})$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Then Prox_ϕ is increasing and:

$$|\text{Prox}_\phi \xi - \text{Prox}_\phi \eta| \leq \frac{|\xi - \eta|}{1 + \theta}$$

for all $\xi, \eta \in \mathbb{R}$, where $\theta = \inf_{\zeta \in \mathbb{R}} \phi'(\zeta)$.

Corollary (Lipschitz Continuity)

The proximity operator Prox_ϕ is Lipschitz continuous on $\mathbb{R}$.

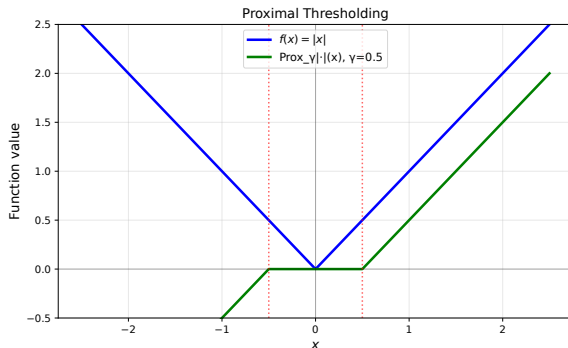
Applications to real-valued functions provide bounds for the rate of convergence.

Proximal Thresholding and Soft Shrinkage

Example (Soft Thresholding on Real Line)

For $\gamma \in \mathbb{R}_{++}$ and $x \in \mathbb{R}$:

$$\text{Prox}_{\gamma\|\cdot\|} x = \begin{cases} \left(1 - \frac{\gamma}{|x|}\right) x & \text{if } |x| > \gamma \\ 0 & \text{if } |x| \leq \gamma \end{cases}$$



Proximal Thresholding for Signal Processing

Example (Soft Thresholding Signal Denoising)

Given a noisy signal $x \in \mathcal{H}$, the soft-thresholded version with threshold γ is:

$$\text{Prox}_{\gamma\|\cdot\|_1}x = \text{sign}(x) \max\{|x| - \gamma, 0\}$$

- Removes noise by thresholding coefficients
- Preserves signal energy better than hard thresholding
- Used extensively in wavelets and compressed sensing

Application in L^2 Spaces

For $\mathcal{H} = L^2(\Omega, \mathcal{F}, \mu)$ with measure μ :

$$P(\omega) = \text{soft}_{[-\gamma, \gamma]}X(\omega) = \text{sign}(X(\omega)) \max\{|X(\omega)| - \gamma, 0\}$$

Then $P = \text{Prox}_{\gamma f}X$ where $f : \mathcal{H} \rightarrow \mathbb{R}$ is the appropriate convex function.

Proximal Gradient Descent Algorithm

Algorithm: Proximal Gradient Method

Given $f, g \in \Gamma_0(\mathcal{H})$ and $x_0 \in \mathcal{H}$, set $\gamma \in (0, 2/L)$:

- ① Initialize: $x := x_0, t := 0$
- ② Repeat until convergence:
 - ① $x^+ := \text{Prox}_{\gamma f}(x - \gamma \nabla g(x))$
 - ② $x := x^+, t := t + 1$

```
# Example: minimize ||Ax - b||^2/2 + gamma * ||x||_1
import numpy as np
from scipy.optimize import proximal_gradient_method

def soft_threshold(x, threshold):
    return np.sign(x) * np.maximum(np.abs(x) - threshold, 0)

def proximal_l1(x, gamma):
    return soft_threshold(x, gamma)

# Forward-backward splitting: minimize f(x) + g(x)
# where f is convex, g has L-Lipschitz gradient
```

Convergence: $x_t \rightarrow x^*$ as $t \rightarrow \infty$ under standard conditions.

Python Example: Soft Thresholding

```
import numpy as np
import matplotlib.pyplot as plt

def soft_threshold(x, gamma):
    """Soft thresholding operator (proximal of L1 norm)"""
    return np.sign(x) * np.maximum(np.abs(x) - gamma, 0)

# Example: Denoising
np.random.seed(42)
signal = np.array([1, 2, 3, 2, 1])
noise = np.random.normal(0, 0.5, len(signal))
noisy_signal = signal + noise

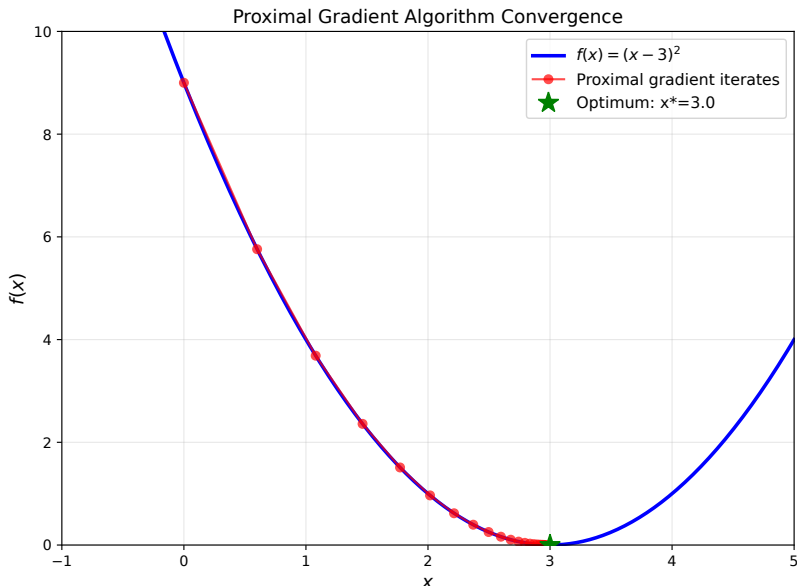
# Apply soft thresholding with gamma=0.5
gamma = 0.5
denoised = soft_threshold(noisy_signal, gamma)

print(f"Original:    {signal}")
print(f"Noisy:        {noisy_signal}")
print(f"Denoised:     {denoised}")

# Verify: check subdifferential condition
# For  $f(x) = \gamma * ||x||_1$ :
#  $\text{Prox}_f(x) = \text{soft\_threshold}(x, \gamma)$ 
```

Output illustrates how proximity operators remove noise while preserving structure.

Convergence Analysis



Perspective Functions

Definition (Perspective Function)

For $\varphi \in \Gamma_0(\mathcal{H})$ with $\text{dom}\varphi^* = \mathcal{H}$, define:

$$f : \mathbb{R} \times \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}, \quad (\xi, x) \mapsto \begin{cases} \xi\varphi(x/\xi) & \text{if } \xi > 0 \\ (\text{rec}\varphi)(x) & \text{if } \xi = 0 \\ +\infty & \text{if } \xi < 0 \end{cases}$$

Theorem (Proximity of Perspective Functions)

Let $(\xi, x) \in \mathbb{R} \times \mathcal{H}$ and $\gamma \in \mathbb{R}_{++}$. Then $f \in \Gamma_0(\mathbb{R} \oplus \mathcal{H})$ and:

If $\xi + \gamma\varphi^*(x/\gamma) \leq 0$: $\text{Prox}_{\gamma f}(\xi, x) = (0, 0)$.

Applications in regularized optimization and hyperbolic constraints.

Proximal Thresholders

Theorem (Characterization of Proximal Thresholders)

Let Ω be a nonempty closed interval in $\mathbb{R}$ and $\phi \in \Gamma_0(\mathbb{R})$. Then the following are equivalent:

- ① Prox_ϕ is a proximal thresholder on Ω
- ② $\phi = \psi \oplus \sigma_\Omega$, where $\psi \in \Gamma_0(\mathbb{R})$ is differentiable at 0 with $\psi'(0) = 0$

Example (Soft Threshold)

The soft thresholding operator (Example 24.20) is a proximal thresholder on $[-\gamma, \gamma]$.

Proximal thresholders provide a systematic way to understand and design operators for signal processing and optimization.

Moreau Envelope

Definition (Moreau Envelope)

The **Moreau envelope** of $f \in \Gamma_0(\mathcal{H})$ with parameter $\gamma > 0$ is:

$$f^\gamma(x) := \inf_{p \in \mathcal{H}} \left(f(p) + \frac{1}{2\gamma} \|p - x\|^2 \right)$$

- ① f^γ is convex and continuously differentiable
- ② $\nabla f^\gamma(x) = \frac{1}{\gamma}(x - \text{Prox}_{\gamma f}(x))$
- ③ $f^\gamma(x) = \inf_p (f(p) + \frac{1}{2\gamma} \|p - x\|^2)$, with $p^* = \text{Prox}_{\gamma f}(x)$

The Moreau envelope smooths f and is fundamental to optimization algorithms.

Properties of Moreau Envelope

Theorem (Derivative of Moreau Envelope)

For $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \mathbb{R}_{++}$:

$$(\forall x \in \mathcal{H}) \quad \nabla f^\gamma(x) = \gamma^{-1}(x - \text{Prox}_{\gamma f}(x))$$

Corollary (Gradient Equivalence)

The proximity operator can be recovered from:

$$\text{Prox}_{\gamma f}(x) = x - \gamma \nabla f^\gamma(x)$$

The Moreau envelope is used in:

- Smoothing nonsmooth functions for optimization
- Deriving algorithms with gradient-based structure
- Analyzing convergence rates

Lasso and Sparse Regression

Problem (Lasso Problem)

Minimize:

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$.

Solution (Proximal Gradient Method)

Iterate:

$$x_{k+1} = \text{Prox}_{\gamma\lambda\|\cdot\|_1}(x_k - \gamma A^T(Ax_k - b))$$

where $\gamma \in (0, 2/\|A\|^2)$.

The proximity operator of $\gamma\lambda\|\cdot\|_1$ is soft thresholding:

$$\text{Prox}_{\gamma\lambda\|\cdot\|_1}(z) = \text{sign}(z) \max\{|z| - \gamma\lambda, 0\}$$

Matrix Completion and Nuclear Norm Minimization

Problem (Matrix Completion)

Minimize:

$$f(M) = \frac{1}{2} \|P_{\Omega}(M - M^*)\|_F^2 + \lambda \|M\|_*$$

where $\|M\|_*$ is the nuclear norm (sum of singular values).

Solution (Proximal Operator of Nuclear Norm)

For $M \in \mathbb{R}^{m \times n}$ with SVD $M = U\Sigma V^T$:

$$\text{Prox}_{\gamma\|\cdot\|_*}(M) = U\Sigma_{\gamma}V^T$$

where $\Sigma_{\gamma} = \text{Diag}(\text{Prox}_{\gamma}(\sigma_1), \dots, \text{Prox}_{\gamma}(\sigma_r), 0, \dots, 0)$.

This reduces to soft thresholding of singular values.

Proximal Splitting Algorithms

Algorithm: Douglas-Rachford Splitting

To minimize $f(x) + g(x)$ where $f, g \in \Gamma_0(\mathcal{H})$:

- 1 Initialize: $y_0 \in \mathcal{H}$
- 2 Repeat: $x_{k+1} = \text{Prox}_f(y_k)$, $y_{k+1} = y_k + \text{Prox}_g(2x_{k+1} - y_k) - x_{k+1}$

Converges to a minimizer of $f + g$ under standard conditions.

```
# Douglas-Rachford splitting
def douglas_rachford(f_prox, g_prox, y0, max_iter=100):
    y = y0
    for k in range(max_iter):
        x = f_prox(y)           # Prox_f
        g_eval = g_prox(2*x - y) # Prox_g
        y = y + g_eval - x
    return x
```

Key advantage: Handles cases where minimizing $f + g$ directly is difficult.

Proximity Operators on Matrix Spaces

Theorem (Singular Value Thresholding)

Let $\phi \in \Gamma_0(\mathbb{R})$ be even and $A \in \mathbb{R}^{M \times N}$ with SVD $A = U\Sigma V^T$. Then:

$$\text{Prox}_\phi(A) = U\Sigma_\phi V^T$$

where $\Sigma_\phi = \text{Diag}(\text{Prox}_\phi(\sigma_1), \dots, \text{Prox}_\phi(\sigma_r), 0, \dots, 0)$.

Example (Nuclear Norm Proximity)

For the nuclear norm $\|A\|_* = \sum_i \sigma_i$:

$$\text{Prox}_{\gamma\|\cdot\|_*}(A) = U\Sigma_\gamma V^T$$

where $\Sigma_\gamma = \text{Diag}(\max\{\sigma_1 - \gamma, 0\}, \dots)$ (soft threshold on singular values).

Applications:

- Matrix completion

Summary: Core Concepts

- **Proximity Operator:** Balances minimizing a convex function f with staying close to a point x
- **Characterization:** $p = \text{Prox}_{\gamma f}(x) \iff x - p \in \gamma \partial f(p)$
- **Key Property:** Single-valued, nonexpansive operator with strong convergence properties
- **Special Cases:**
 - 1 Projections onto convex sets
 - 2 Soft thresholding (for ℓ_1 norm)
 - 3 Singular value thresholding (for nuclear norm)
- **Moreau Envelope:** $f^\gamma(x) = \inf_p (f(p) + \frac{1}{2\gamma} \|p - x\|^2)$ provides smoothing
- **Algorithms:** Proximal gradient method, Douglas-Rachford splitting, ADMM

Key Results

- 1 Proximity operators are resolvent operators of the subdifferential
- 2 They inherit important properties:
 - Weak sequential continuity
 - Nonexpansiveness: $\|\text{Prox}_f x - \text{Prox}_f y\| \leq \|x - y\|$
 - Fixed points correspond to minimizers of f
- 3 Composition rules enable flexible operator design
- 4 Explicit formulas available for:
 - Quadratic functions
 - Norms (ℓ_1 , nuclear norm, etc.)
 - Indicators of convex sets
 - Distance functions
- 5 Moreau envelope: Key tool for smoothing and algorithm design
- 6 Proximal algorithms (FISTA, ADMM, etc.) are among the most efficient modern methods

Applications Across Domains

Signal Processing

- Denoising via soft thresholding
- Compressed sensing
- Wavelet shrinkage

Optimization and Machine Learning

- Lasso and sparse regression
- Matrix completion
- Robust PCA
- Convex optimization algorithms

Statistical Learning

- Regularized regression
- Feature selection
- Smoothing in high dimensions

Further Reading I

Key references from Chapter 24:

- Proposition 24.1-24.30: Core theory of proximity operators
- Theorem 24.52: Proximal thresholds characterization
- Examples 24.1-24.66: Extensive worked examples
- Sections 24.4-24.8: Applications and special cases

For practical algorithm implementations:

- Proximal gradient method (forward-backward splitting)
- Douglas-Rachford splitting
- ADMM (Alternating Direction Method of Multipliers)
- FISTA (Fast Iterative Shrinkage-Thresholding Algorithm)

Modern optimization frameworks:

- ProximalAlgorithms.jl (Julia)
- CVX (MATLAB)
- Scikit-optimize (Python)